

Korea Expression Archive: A public repository for functional genomics data

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Introduction

In recent years, large-scale genomic data have become the cornerstone of modern biomedical and life science research. The increasing availability of multi-omics data has enabled data-driven research to elucidate complex biological mechanisms and support biomarker discovery. Over the past few decades, microarray and high-throughput sequencing (HTS) technologies have played a pivotal role in advancing functional genomics by enabling large-scale gene expression profiling and molecular characterization of diverse biological conditions. However, a substantial number of datasets remain scattered across individual laboratories or unpublished, limiting their accessibility and reuse. Therefore, establishing a well-curated public repository is essential to facilitate the use of underutilized data and support the growing demand for multi-omics research.

The Korea Expression Archive (KEA; <https://kbds.re.kr/KEA>) was developed as an expanded platform building upon KArray, a public repository for microarray data. While KArray primarily focused on microarray datasets, KEA extends its scope to processed data derived from HTS experiments. KEA offers data submission, search, browsing, and download functionalities to promote data sharing and enhance data accessibility.

Main

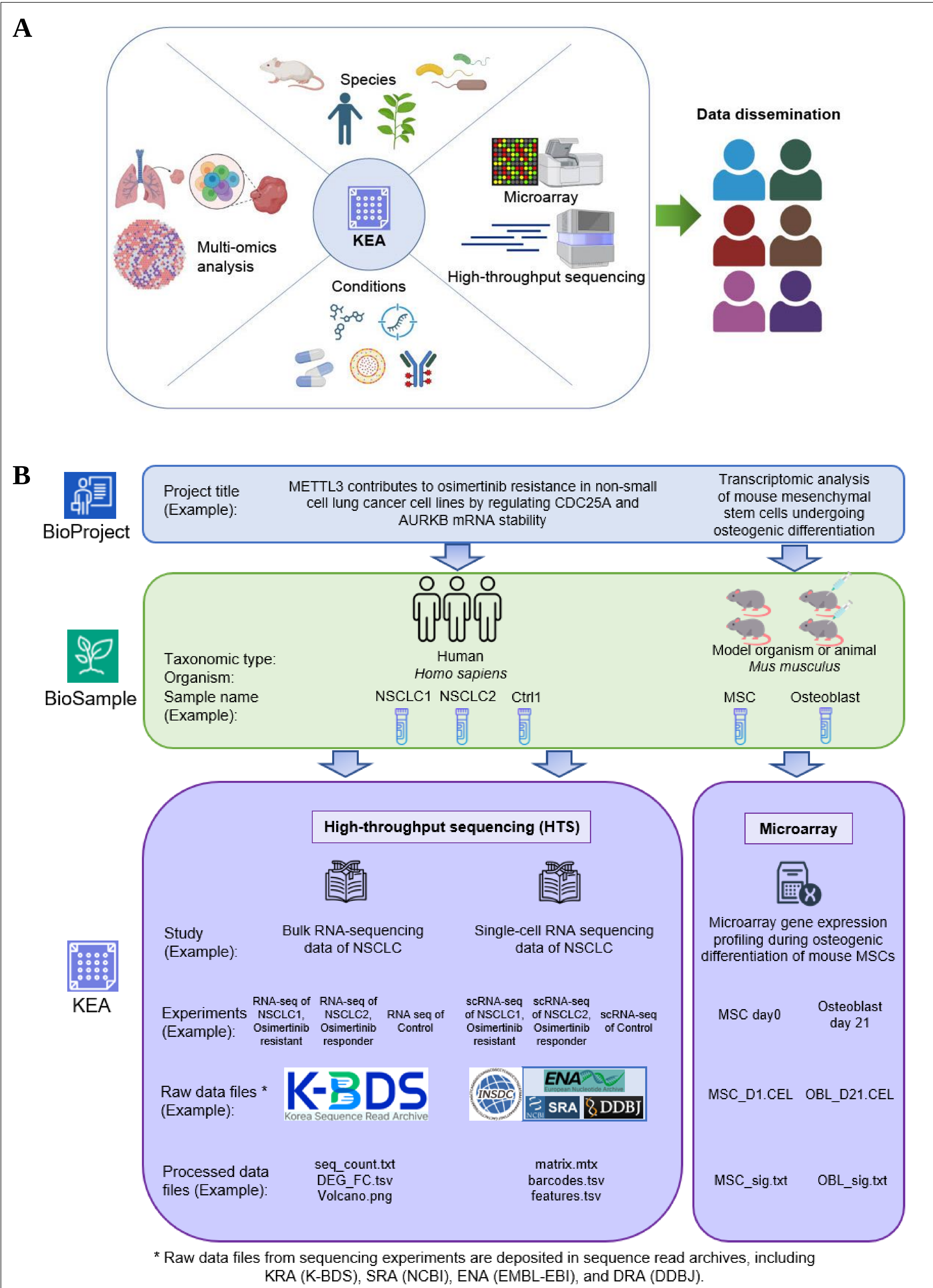


Figure 1. Overview of KEA

(A) Schematic illustration of KEA. KEA integrates, curates, and disseminates microarray data and processed HTS data across diverse species and experimental conditions to support data reuse for multi-omics analyses. The illustration was created with BioRender (<https://www.biorender.com/>).

(B) The database structure of KEA. BioProject defines the overview of a study, while BioSample specifies the organism and sample attributes. Within KEA, data are expanded into detailed studies and experimental information, including data files.

HTS, high-throughput sequencing.

Submission process

The KEA submission process provides a streamlined workflow for submitting and integrating complex multi-omics data. The workflow is structured to:

BioProject: Every submission begins with a BioProject that contains comprehensive research information, including the project title and description.

BioSample: For most data types, a BioSample is registered to describe the biological source material, with a comprehensive Taxonomy section for accurate classification.

KEA Data: The final processed data files derived from raw data are submitted and linked to the corresponding BioProject and BioSample. For microarray data, both raw and processed data are required for submission.

Main

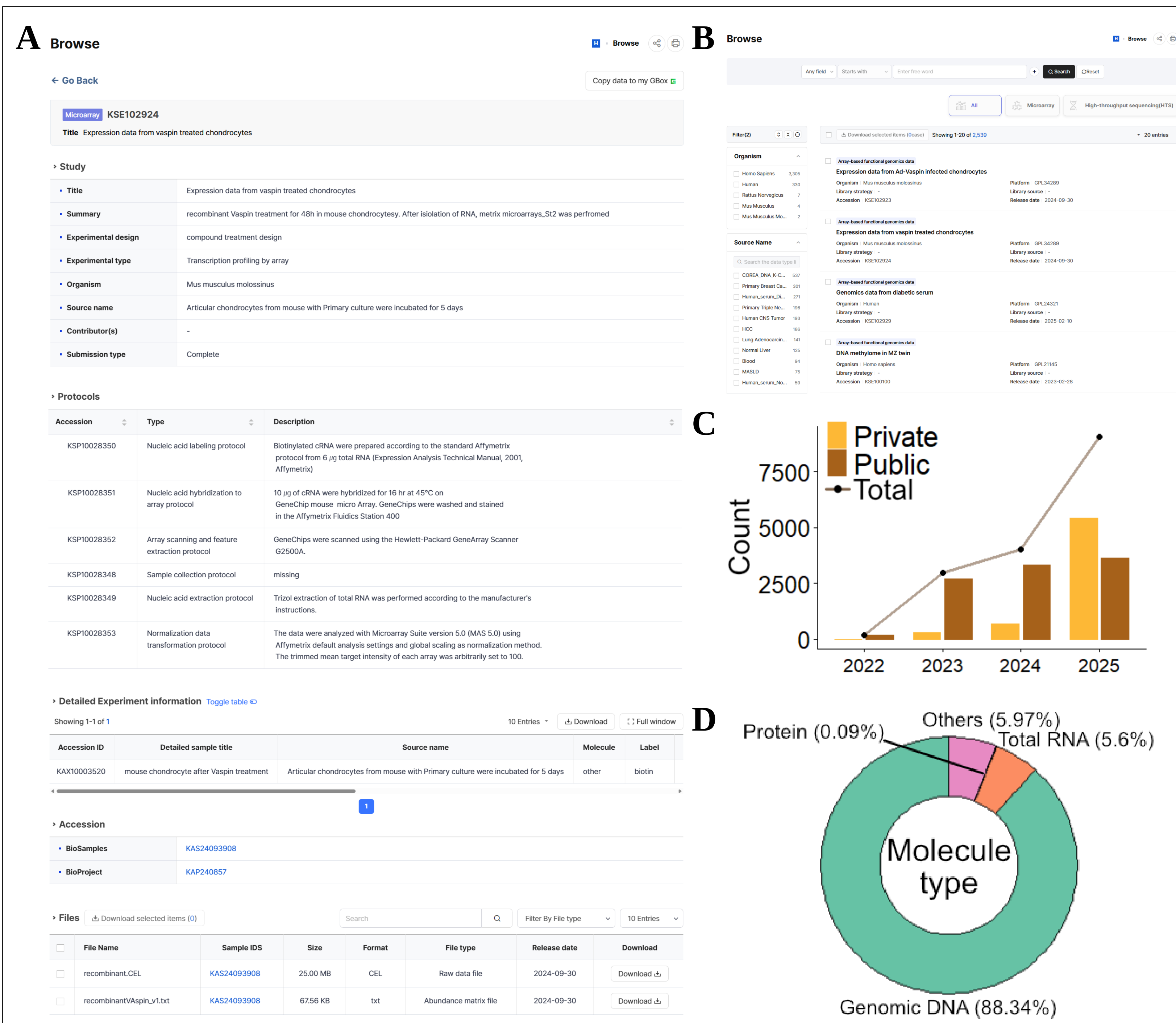


Figure 2. Web interface and statistical summary of KEA

(A) The detailed view page shows dataset information, including the study description, experimental and sample attributes, and data files.

(B) The browse page for data retrieval with search and filtering functions by organism and source name. Each data point is displayed with associated metadata, including organism, platform, accession number, and submission date. (C) Bar plot representing the number of Experiments. Each bar is divided into public and private records, while the dotted line indicates the cumulative total number of Experiments across years.

(D) The donut chart shows the proportion of genomic DNA, total RNA, protein, and other molecule types, with genomic DNA accounting for the largest share.

Conclusions and future directions

KEA addresses the need for a centralized and well-curated repository for sharing and disseminating data. KEA ensures consistent data quality and interoperability with global resources by adopting metadata standards aligned with the Minimum Information About a Microarray Experiment (MIAME) and the Minimum Information About a Next-generation Sequencing Experiment (MINSEQE) guidelines, which are internationally standardized formats for microarray and HTS data.

KEA offers a user-friendly web interface for data submission, retrieval, and reuse. As a public repository within the Korea BioData Station (K-BDS; <https://kbds.re.kr/>), KEA contributes to facilitating data-driven and integrative multi-omics research.

Moving forward, we plan to broaden KEA's scope by incorporating web-based data analysis functionalities that allow users to explore and analyze submitted data directly within the platform without requiring local computational resources. Additionally, KEA seeks to establish collaborative frameworks with other international functional genomics databases to promote global data exchange. Through continuous development, KEA aims to evolve into a comprehensive functional genomics data platform that facilitates the reuse of data to advance genomic and transcriptomic studies.

References

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